AIORI HACKATHON 3 | PS-019 | TC-038

PrivacyTwin

An Evidence-Centric Personal-Data Discovery Platform with ReviewGuard for Versioned, Contradiction-Aware Human Decisions

DETECT → EXPLAIN → CHALLENGE → REVIEW → TRACK CHANGE

Proposed by: Team Cognitive Crew
Nivedita V P, Divya A, Bharani H(Mentor)

Submission proposal • October 2026

Challenge: Personal-Data Discovery and Privacy Evidence Inventory

Abstract / Problem Description

AIORI 3 • PS-019: Personal-Data Discovery and Privacy Evidence Inventory

ABSTRACT

PrivacyTwin is a proposed local-first platform for discovering candidate personal-data fields in CSV and SQLite sources. It combines deterministic signals with bounded Gemini-powered agents to explain ambiguous fields, preserve competing interpretations, and identify missing purpose or retention evidence. ReviewGuard records versioned human decisions and selectively marks a decision stale when a dependency it relied on changes. The system produces a field-level Evidence Passport, review history, inventory and gap report. Its output is technical evidence for human review—not a legal-compliance determination.

PROBLEM

- Misleading or domain-specific column names make personal-data classification uncertain.
- Pattern-only rules can confuse personal identifiers with business identifiers or substrings inside other identifiers.
- Single confidence scores may hide competing evidence and mixed-value columns.
- Previous approvals can become stale after data, rules or metadata change.
- Reviewing every field again is expensive; silently retaining obsolete decisions is risky.

Background

Existing approaches solve parts of the problem; PrivacyTwin connects evidence to review freshness.

Common approaches

- Regex and configurable rules
- ML / semantic classification
- Privacy knowledge representation
- Metadata catalogues and data lineage
- Policy and runtime requirement systems

PrivacyTwin approach

- Combine technical evidence signals
- Preserve supporting and contradicting evidence
- Use Gemini for bounded contextual explanation
- Persist scoped human decisions in ReviewGuard
- Reopen only contracts affected by relevant changes

Key distinction

From “What class is this field?” to “What evidence supports this decision, what conflicts with it, and can we still trust the last review?”

Literature & RFC Review

Focused comparison of related research areas; verify full bibliographic details before final submission.

Work / area	Existing contribution	PrivacyTwin distinction
TEIRESIAS	Scalable personal-data discovery and inventory	Emphasis here is contradiction-aware evidence and review freshness
PRIVAFRAME	Knowledge-graph representation and semantic sensitive-data detection	PrivacyTwin focuses on field evidence cases and scoped review decisions
PII in Data Lakes	PII discovery with metadata lifecycle management	Adds dependency-aware reopening and explicit review freshness
Column-level lineage	Data lineage and audit trails	Focuses on whether a classification/metadata decision remains valid
PrivGraph	Privacy relationships and aggregation risks	Different objective from operational field discovery and review tracking
Multi-engine PII classification	Combines classification engines and confidence	Adds competing evidence, scoped contracts and stale-review testing
PAM / privacy requirements	Specifying privacy requirements and runtime validation	PrivacyTwin reports technical evidence; it does not enforce policy or infer lawful basis

Research gap

Evidence-centric classification combined with dependency-aware, selective reopening of human review decisions.

Proposed Solution

PrivacyTwin combines detection, bounded Gemini assistance, metadata evidence and ReviewGuard.



Output: Evidence Passport • Inventory • Review history • Checklist • gap_report.csv

- Five classes: Identity, Contact, Location, Government/Official ID, Demographic.
- Gemini agents suggest contextual interpretations and explanations; local validators verify evidence references.
- ReviewGuard uses deterministic dependency checks—not an LLM—to decide whether a review contract becomes stale.
- Metadata Fix Preview shows proposed gap closure without saving until the user explicitly confirms.

Solution Optimization

Before-and-after comparison of the proposed improvements.

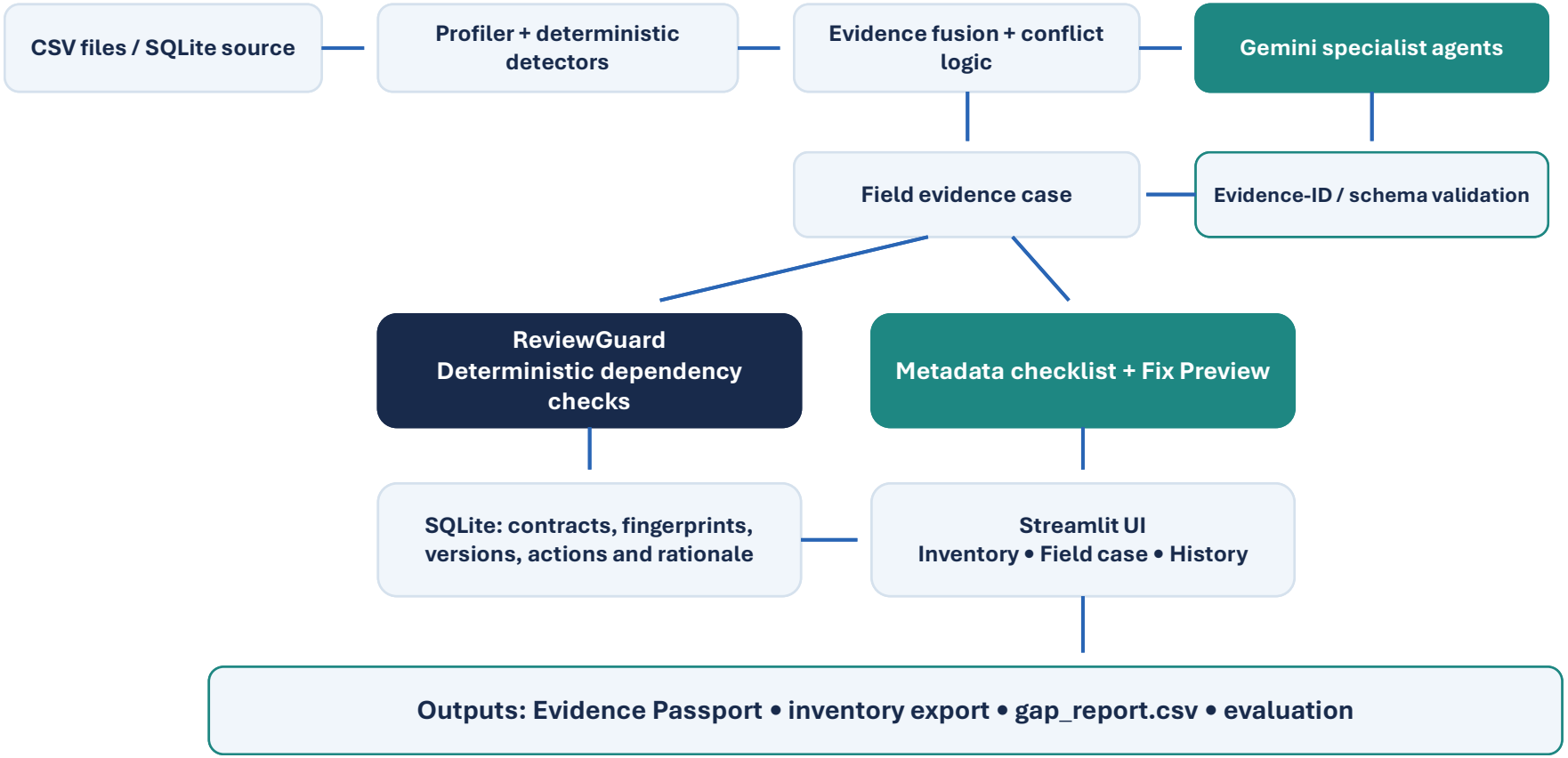
Before	After with PrivacyTwin
One label and a confidence score	Supporting + contradicting evidence, confidence and ambiguity
Rules may confuse embedded identifier patterns	India-first validators plus deceptive negative cases and grouped evidence
Review decisions are not persisted or scoped	Versioned classification and metadata review contracts
Every rescan may prompt repeat review	Selective stale marking based on changed dependencies
Metadata gaps are manually inspected	Configurable checklist, Fix Preview and gap_report.csv
A proposed change may be saved prematurely	Preview is non-mutating; save requires explicit action
LLM output could be treated as authoritative	Gemini suggestions are validated; deterministic code owns review-state changes

Optimization target

Reduce unnecessary repeat reviews while retaining transparent evidence and review integrity.

Architecture Design (UML / Component View)

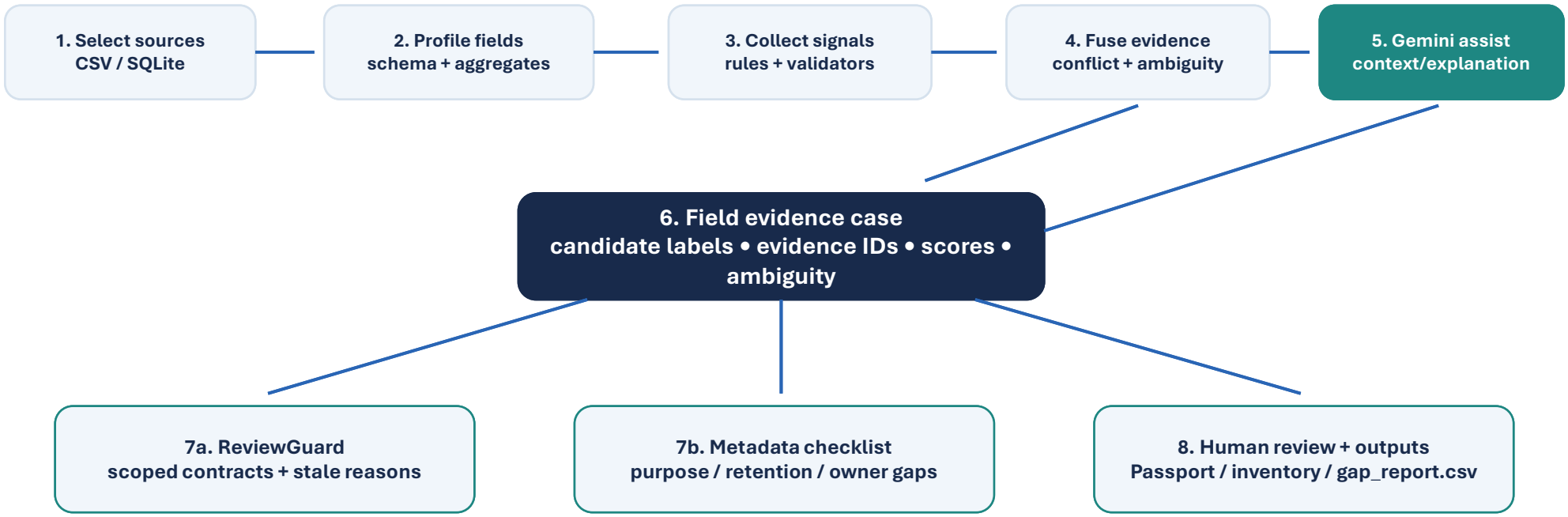
Recommended hackathon build: one Streamlit process, modular Python core, SQLite persistence and optional Gemini API.



Gemini API key is read from GEMINI_API_KEY; never embed it in source code or browser UI.

Data Flow Diagram (DFD)

Data flows from local sources to field-level evidence cases and reviewable outputs.



Privacy safeguard: use synthetic records, mask examples, minimize Gemini payloads, and never send complete datasets by default.

Deployment Timeline

Implementation timeline: 16th Oct – 31st Oct; milestones describe planned work.

16th–18th Oct	**Foundation**	Local CSV/SQLite scan, profiler, synthetic data fixtures, repository setup
19th–21st Oct	**Detection**	Versioned rules, India-first validators, grouped evidence, contradiction logic
22nd–24th Oct	**ReviewGuard**	Persisted contracts, fingerprints, dependency comparison, stale reasons
25th–28th Oct	**Metadata + UI**	Checklist, Fix Preview, Evidence Passport, review history and exports
29th–31st Oct	**Evaluation + demo**	Held-out metrics, 15–20 change tests, mentor demo and backup recording

Day-one checklist Confirm repository access, proposal template, licensing/attribution and submission requirements early.

Bibliography

Selected related works identified in the current literature review. Verify final citation metadata before submission.

- [1] Grünewald and Schurbert. “TEIRESIAS: Scalable Discovery and Continuous Inventory of Personal Data at Rest in Cloud Native Systems.” ICSOC, 2022.
- [2] Gambarelli and Gangemi. “PRIVAFRAME: A Frame-Based Knowledge Graph for Sensitive Personal Data.” Big Data and Cognitive Computing, 2022.
- [3] “Managing Personal Identifiable Information in Data Lakes.” Research on PII discovery, metadata models and lifecycle management.
- [4] “Automated Column-Level Data Lineage and Audit Trails for GDPR Compliance in MarTech.”
- [5] Zuo and Lee. “Addressing Memorization and Aggregation Risks in AI: A Knowledge Graph Approach to Privacy.” Applied Sciences, 2026.
- [6] Sethi et al. “Classifying 1.5 Million Columns: An Automated Multi-Engine Approach to Enterprise PII Governance.” ICOSAAS, 2026. DOI: 10.1109/ICOSAAS68663.2026.11648896.
- [7] Shahriar et al. “A Comparative Study on Textual and Multimodal PII Detection Using Large Language Models.” Computer Standards & Interfaces, 2026.
- [8] Pantelelis and Kalloniatis. “PAM: A domain-specific language for specifying privacy requirements from regulation to runtime.” Requirements Engineering, 2026.
- [9] User-provided source: NiviAkka-DRDO-Hackathon.pdf. PrivacyTwin design recommendations, ReviewGuard, test scenarios and implementation boundaries.
- [10] Google. Gemini API documentation. Consult the current official documentation for SDK, model availability, security and usage limits.

Thank You

PrivacyTwin

Discover evidence. Explain uncertainty. Reopen only what changed.

Questions & Discussion

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